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Non-invasive blood glucose level measurement method under the consideration of various physiological traits using machine learning

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Abstract. Physiological traits such as skin colour, body fat, skin thickness, pH value, hydration level, and others play a crucial role in non-invasive glucose monitoring as they directly impact the performance of blood glucose level measurement (BGLM). The DC component of the photoplethysmogram (PPG) signal accounts for a significant portion of BGLM error caused by these physiological traits. This paper proposes a novel method to estimate the error correction factor (ECF) by correlating the DC component with BGM error. Linear Regression (LR) and Support Vector Machine (SVM) regression models are employed to predict blood glucose levels (BGL) and estimate the ECF. The proposed method uses PPG records from 15 healthy subjects for model training and 30 additional records for validation. Testing across diverse age groups demonstrated the value of Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2) as ±5.1575 mg/dL and 0.9581, respectively. Compared to existing work, the proposed method brings a reduction of 30.83% in RMSE and an improvement of 2.13% in R^2. Clarke Error Grid Analysis confirmed that 99% of predictions fell within clinically accurate Zone A. The proposed method offers significant advantages, including reduced cost, simplicity, comfort, and suitability for wearable real-time BGM systems.

Keywords. Non-Invasive; PPG; BGLM; NIR sensor; ECF; SVM.

1. Introduction

Diabetes, a chronic metabolic condition continues to affect an ever-growing number of people globally. The global burden of diabetes has experienced unprecedented growth, escalating from approximately 200 million cases in 1990 to over 830 million in 2022, representing a more than four-fold increase that particularly affects low- and middle-income nations [1, 2]. Between 2000 and 2019, there was a 3% increase in diabetes mortality rates by age [2]. Diabetes, while not directly fatal, is associated with serious complications such as cardiovascular disease, kidney failure, and nerve damage, which can lead to life-threatening consequences if left unmanaged [3–5]. Managing blood glucose levels (BGLs) remains a crucial part of controlling diabetes, preventing these complications from arising. Traditional self-monitoring of blood glucose (SMBG) methods, while clinically proven and widely adopted, impose significant psychological burdens on patients. The invasive approaches necessitate multiple daily finger-prick tests, leading to pain, potential infection risks, and compromised patient compliance that can ultimately result in suboptimal diabetes management [4–8]. The invasive nature of conventional glucose monitoring creates a paradoxical situation where the very tool designed to improve patient health becomes a source of discomfort and anxiety, thereby potentially undermining treatment adherence and long-term outcomes [9, 10].

The quest for non-invasive glucose monitoring represents more than a technological advancement; it embodies a paradigm shift toward patient-centered healthcare that prioritizes comfort, accessibility, and continuous monitoring capabilities. Recent technological breakthroughs have positioned several non-invasive approaches at the threshold of clinical viability, with some achieving regulatory approval and commercial deployment [3, 11, 12]. These developments signal a transformative moment in diabetes care, where the convergence of optical technologies, artificial intelligence, and advanced materials science promises to revolutionize how patients interact with their glucose monitoring systems. Thus, a shift towards non-invasive glucose monitoring has captured significant research interest, as it aims to reduce patient discomfort and improve the overall ease of diabetes management.

The landscape of non-invasive glucose monitoring encompasses three primary technological domains: optical

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methods, electromagnetic sensing approaches, and advanced biosensor technologies [13–15]. Optical methods include near-infrared (NIR)/mid-infrared (MIR) spectroscopy, Raman spectroscopy, photoacoustics, optical polarimetry, and photoplethysmography (PPG). Electromagnetic approaches use radio/ microwave /terahertz sensors that probe dielectric properties of tissue [6, 14]. Biosensor-based methods measure glucose-related analytes in alternative biofluids (sweat, tears, saliva) via enzymatic or electrochemical sensors [6, 15]. Each category has trade-offs in penetration depth, specificity, and practicality. Each category offers distinct advantages and faces unique technical challenges that influence their clinical applicability and commercial viability. Recent surveys note that optical techniques (especially PPG and NIR) dominate published non-invasive blood glucose measurement (NIBGM) work ($\approx$ 58% of studies) while electrochemical sensors (sweat, tears) and mixed methods are less common [6, 9, 13, 16]. Despite significant technological advances, non-invasive glucose monitoring continues to face substantial challenges that impede widespread clinical adoption and limit measurement accuracy. These challenges span technical, physiological, implementation, and regulatory domains, each requiring innovative solutions and continued research investment. The use of AC component is much more common in BGM and most systems do not consider the DC component of PPG, which is sensitive to physiological traits [17]. The DC component reflects the constant absorption of light by non-pulsatile tissues and provides critical information about absorption by tissue composition of body skin [17, 18]. Wearable systems often lack real-time accuracy under varying skin and body conditions [13, 19]. Hardware complexity, calibration drift, biological variability, need for miniaturized, and cost remain barriers to broader adoption.

The proposed work addresses these challenges by proposing a simple, low-cost, and real-time NIBGM system using a NIR sensor and NodeMCU-based architecture. The proposed work uses a single 940nm wavelength NIR sensor. The proposed method introduces a novel error correction factor (ECF) based on the DC component of PPG signal that is used to minimize physiological influences on measurement accuracy. The key contribution of the proposed work includes:

- *Error Correction Factor (ECF)*: The AC component consideration of the PPG signal is the conventional approach to predict BG level while mostly DC component is not considered, which is sensitive to physiological traits. The utilization of DC component is the key idea of the proposed work. A novel ECF is introduced to directly modeling physiological influences on measurement accuracy that is based on the DC component of PPG signal.
- *Simple hardware and Machine Learning (ML) Techniques*: Rather than the multi-wavelength technique and complex hardware, the proposed work used a single 940nm wavelength to generate PPG signal that is used for glucose measurement and predict physiological influences. After extracting AC and DC component form the PPG signal; rather than the sophisticated ML Techniques, the basic LR and SVM techniques are used to estimate BGL and ECF from these components respectively. This is the simplest hardware configuration having low cost as well as less complex ML algorithms with great accuracy as per the best of our knowledge.

The remainder of this paper is structured as follows. Section 2 briefly reviews recent works using PPG for the estimation of blood glucose. The details of methodology including basic theory, NIR sensor deployment & implementation, measurement site selection, hardware architecture, and experimental setup with protocol has been discussed in the section 3 for proposed work. The blood glucose prediction using machine learning and related performance parameters has been discussed in the section 4. Section 5 elaborates result validation and discussions, and section 6 concludes the research and future scope.

2. Related work

This section reviews recent advancements in NIBGM systems, organizing prior work according to optical methodology, algorithmic approach, and evolving model complexity, before outlining avenues for future research.

Early optical spectroscopic approaches to NIBGM rely on either single-wavelength or multi-wavelength interrogation of tissue in the visible, NIR and MIR bands, often combined with PPG. Single- wavelength NIR methods have been widely explored: Hexa *et al* [20] employed a 940nm sensor to estimate blood glucose, as did Yadav *et al* [21] and Jain *et al* [22]. In contrast, Joshi *et al* [23] demonstrated that dual- wavelength sensing (940nm and 1300nm) can enhance prediction accuracy by probing different tissue absorption characteristics. Extending this concept further, Castro *et al* [24] and Li *et al* [25] integrated three or more NIR wavelengths, showing that multi-wavelength analysis can compensate for individual spectral interferences and yield more robust glucose estimates. Beyond NIR/MIR sensors, recent studies have explored visible-light PPG using smartphone cameras: Al-Dhaheri *et al* [26] used systolic-peak heights in reflectance mode under ambient illumination, achieving an RMSE of 10.44 mg/dL, while Islam *et al* [27] extracted five temporal and amplitude features from smartphone PPG and applied PCR, PLS and SVR to keep prediction errors below 17.02 mg/dL. Taken together, these single- versus multi- wavelength optical approaches illustrate a clear trade-off: single wavelengths offer simplicity and low hardware cost, where- as multi-wavelength systems improve spectral discrimination at the expense of added sensor complexity.

The algorithmic strategies for NIBGM can similarly be grouped into those that do not employ machine learning

(ML) and those that do. Early non-ML methods typically utilized straightforward statistical or regression techniques: simple linear regression on PPG features yielded moderate accuracy in Al-Dhaheri's work [27], and Owoeye *et al*'s [28] Arduino-based 940nm system achieved R^2 = 0.9369 using a univariate model. Smartphone-based PPG studies also applied PCR, PLS and SVR without extensive feature engineering, demonstrating feasibility for consumer-grade platforms [27]. Javid *et al* [29] compared transmittance and reflectance modes, noting a systematic voltage response change with glucose and reporting an average error of 8.27%. In contrast, ML-based approaches have proliferated over the past decade, leveraging both classical and advanced learning algorithms. Monte-More- no *et al* [30] applied ensemble ML on PPG and demographic inputs to meet clinical blood pressure standards and detect hyperglycemia with only 1.9% missed abnormal cases. Prabha *et al* [31] tested SVM and extreme gradient boosting regression (XGBR) on wristband-derived signals from 217 subjects, finding that XGBR on the five most informative features yielded the highest concordance with reference glucose values. Lee *et al* [32] introduced the Centralized State Sensing (CSS) algorithm for multi-sensor fusion, pairing it with a deep network to achieve 84.29% accuracy on a 349-sample dataset. Subsequent work by Golap *et al* [33] used genetic algorithms to select 46 PPG features for a custom regression model, while Habbu *et al* [34] demonstrated that a simple multilayer perceptron (MLP) could place 80.6% of predictions in Clarke's Zone A. Islam's [27] smartphone studies, though non-ML in their regression phase, laid the groundwork for device-based algorithms. More recent innovations include PSO-MKL-SVR for multi-kernel learning to mitigate confounding factors [35], Empirical Mode Decomposition (EMD) features with CatBoost achieving R^2 = 0.92 [36], and Ensemble Extreme Learning Machine (EELM) applied to eight-wavelength PPG to predict hemoglobin (Hb) with RMSE = 1.72 g/dL [37]. Collectively, these works illustrate how the incorporation of ML transforms NIBGM from proof-of-concept to clinically relevant accuracy.

As ML methods have become more sophisticated, predictive accuracy has steadily improved. Beginning with simple linear regression [26, 28], which achieved RMSEs around 10–18mg/dL, researchers progressed to multivariate PCR and PLS models [27] that reduced error by capturing correlated feature variance. Support vector regression (SVR) further lowered prediction error margins, demonstrating sub-17mg/dL RMSE in smartphone PPG [27]. Transitioning to nonlinear learners, SVM and XGBR exploited complex feature interactions: Prabha's [31] XGBR model, streamlined to five key inputs, achieved state-of-the-art concordance. MLPs provided a bridge to deep learning, with Habbu's [34] network placing 80.6% of predictions in the clinically safe Zone A. ANN architectures incorporating multiple sensor streams pushed R^2 above 0.94, as shown by Yadav *et al* [21]. Genetic-algorithm- based feature selection combined with tailored regressors by Golap *et al* [33] improved robustness over handcrafted features. PSO-MKL-SVR optimized kernel combinations to achi- eve $R^2 > 0.997$ and RMSE = 0.00728 [35], albeit on small datas- ets. Ensemble methods like CatBoost on EMD features attained an Mean Absolute Error (MAE) of 8.01mg/dL [36]. EELM demonstrated that extremely fast training could still secure moderate Hb prediction (R^2 = 0.5776) [37]. Deep convolutional and recurrent architectures such as DCC-Net enabled real-time, offline glucose classification with 92% accuracy [38], while Turja's reflective DVP method with modified Beer–Lambert modeling achieved R^2 = 0.923 and RMSE = 0.309 [39]. As ML models become more advanced, from simple linear regression to deep learning, their predictions usually get more accurate, but they also need more data and computing power.

Even with progress, there are still big challenges. Non-invasive systems aren't yet as accurate as traditional methods, especially when people are moving or have changing body conditions. Wearable devices can also be bulky and expensive because they need many sensors and parts. On top of that, many machine learning models require large, well-labeled datasets and lots of fine-tuning, which makes them hard to scale or use in real time.

To improve, future research should aim for better accuracy while keeping devices simple and easy to use. Promising ideas include using fewer light wavelengths to reduce sensor size, applying transfer learning to speed up model training with existing data, and combining basic physical models with lightweight machine learning to cut down on complexity. By improving measurement stability, simplifying hardware, and making data analysis more efficient, the next generation of non-invasive glucose monitors could become reliable and practical for everyday use.

3. Methodology

The basic theory, sensor deployment, site selection, hardware architecture, and experimental setup with protocol has been described below in detail.

3.1 *NIR spectroscopy*

This spectroscopy offers a novel non-invasive approach to assess glucose concentrations. When a beam of light travels through an aqueous solution containing glucose, the dissolved molecules reduce the transmitted light's intensity at particular wavelengths, generating a distinct spectral pattern unique to glucose. This spectrum enables precise determination of glucose concentration. However, applying this method to measure glucose levels through human skin

presents significant challenges. Studies have investigated the potential of analyzing skin tissue through optical methods, but a significant limitation arises from the restricted depth that light can penetrate biological material, which primarily occurs within the NIR spectral region [38, 40]. A key challenge in this approach is that glucose lacks a unique spectral signature detectable in the NIR range. Moreover, achieving precise measurements is hindered by multiple variables, including the diffusion of light within tissues, fluctuations in absorption rates due to temperature changes, and the heterogeneity of dermal properties across individuals [29]. In this research, infrared (IR) sensors are employed to detect blood glucose levels by measuring light intensity in blood vessels. These sensors consist of an emitter and a photoreceptor, as illustrated in figure 1. The infrared sensor identifies incoming infrared pulse patterns and translates them into a continuous voltage output that directly reflects the strength of the detected infrared radiation. This voltage-based data is subsequently analyzed and interpreted by the NodeMCU platform for deeper evaluation [35, 41]

When a beam of infrared (IR) light passes through glucose-containing tissue, the amount of light absorbed correlates with the concentration of glucose and other components such as water, melanin, and hemoglobin [38, 40]. This forms the theoretical basis for modeling light-tissue interactions using well-established physical laws. Planck's law mathematically defines the energy distribution across wavelengths for electromagnetic emissions from a material in thermal equilibrium at a specific temperature. This equation establishes a relationship between the energy of radiation and its frequency [42]:

$$E = h * f \tag{1}$$

Equation 1 describes the energy associated with the light emitted by the LED at frequency f, where h is Planck's constant. This is relevant in BGM because it quantifies the photon energy input to the biological tissue, which determines how the light interacts with glucose molecules at a quantum level [42].

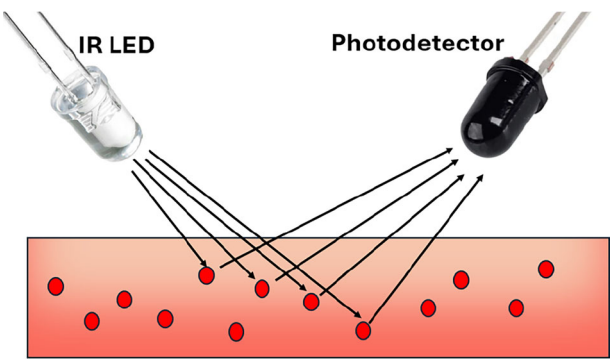


Figure 1. Basic diagram of NIR spectroscopy.

As light propagates through human tissue, it experiences several phenomena, including refraction, absorption, reflection, and scattering. These interactions cause variations in the reflection and refraction of light both within and outside cells and fluids. According to the **Lambert-Beer law**, the amount of light absorbed by a substance depends on its concentration and the path length of the light through it. Although the detection value is theoretically expected to remain constant, minor fluctuations may occur due to changes in glucose molecule concentration.

The relationship is expressed mathematically as [36]:

$$I_r = I_0 \exp(-kcD) \tag{2}$$

where I_r is the detected intensity of light at photo-detector, I_0 denotes the incident light intensity. The parameters k, D, and c denotes molar absorption coefficient, the distance travelled by light, and concentration of the substance, respectively. The multiplication of k and c is replaced by the attenuation coefficient μ (i.e. kc) and the equation (2) can be represented as:

$$I_r = I_0 \exp(-\mu D) \tag{3}$$

The equation (3), can further be expressed as :

$$T_r = \frac{I_r}{I_0} = \exp(-\mu D) \tag{4}$$

where T_r represents transmittance ratio. This ratio is critical in our setup as the photodetector output voltage is directly linked to Ir. The equation (4) can be further be rewritten as:

$$-\log_e(T_r) = \mu D = A \tag{5}$$

where A denotes absorption of light. It defines how much light is absorbed by skin and blood vessels [36, 42]. In the BGL measurement the change in light due to blood and body skin absorption is the sum of glucose and other substances of body. Therefore, overall absorption A_T can be donated as:-

$$A_T = \mu D + \mu_s D = (\mu + \mu_s)D \tag{6}$$

where μ represents attenuation due to blood glucose, and μ_s represents attenuation due to all substances present in blood and body. A_T accounts for attenuation due to both glucose and other substances.In the proposed BGM system the term A_T is essential in estimating the ECF by analyzing to estimate ECF by analyzing the DC component in PPG signal. The DC value inherently contains this combined absorption, and hence the equation (6) models the real-world scenario of overlapping absorption effects in biological tissue.

In the context of the proposed system, the voltage signal detected at the photodetector reflects these absorption relationships. Therefore, the AC and DC components of the PPG signal which are extracted and processed using

regression models, are fundamentally based on the quantitative absorption behavior as described in Equations (1)-(6).

3.2 *Design and development of a non-invasive NIR sensor module*

In previous research, various circuits have been employed by different researchers to interface with data acquisition systems. The fundamental principle behind blood glucose measurement relies on the absorption properties of light in the NIR region. Blood glucose molecules($C_6H_{12}O_6$) capture light because of specific vibrational patterns triggered by overtone and combination effects, leading to photon uptake and molecular movement. These movements stem from the flexibility of covalent bonds, behaving similar to elastic coils that bend and extend. In particular, the CH and OH bonds exhibit elongation and contraction, a phenomenon noted in this spectral range. When the incoming light's frequency aligns with the inherent oscillatory rate of these bonds, the molecules oscillate and take in the light energy [16, 43, 44]. Every sensor module typically includes some common components, as illustrated in figure 2.

Some researchers use only elementary sensor circuits that consist only IR emitter and Photodetector or Photodiode to get PPG Signal [20, 45–48]. The Output of sensor is applied to the analog pin of Arduino Board and convert into digital form by D/A converter of microcontroller. Samples are collected at sample frequency between 100Hz to 125Hz. The collected signal is passed through digital Butterworth filter to pass only the frequency selected frequency range in between 0.15 to 15Hz only. Then the output is sent to Data acquisition system (DAQ). Many researchers use elementary sensor circuits with preamplifier and use digital band pass filter to get quality signal [24, 49]. Some researchers use elementary sensor circuits with preamplifier and use analog band pass filters to get quality signal at the hardware [50, 51]. [29, 37, 38, 52, 53] use commercially available sensor modules to get PPG Signal.

3.3 *Selection of measurement sites*

Non-invasive monitoring of glucose levels in the blood has been investigated across multiple body locations, such as the lips, cheeks, tongue, eyes, earlobes, fingertips, and wrists, as reported in earlier research [54–59]. Among these, the fingertip and wrist are among the most widely adopted sites in practical applications, due to both technical and physiological advantages. From a medical and anatomical perspective, the wrist offers several favorable conditions:

1. It features relatively thin skin and a rich capillary network in the radial and ulnar arterial branches, facilitating effective light penetration and interaction with blood vessels [59].
2. Compared to other sites, the wrist is less prone to external contamination, easier to stabilize, and minimizes discomfort, making it suitable for long-term wearable monitoring [57, 60].
3. NIR light in the 900–1000 nm range can penetrate 2–5 mm into tissue, reaching dermal layers where glucose containing capillary blood resides [28, 37, 54, 61].
4. Moreover, research by Yadav *et al* [21] and Heise *et al* [62] shows that skin thickness and pigmentation at the wrist are sufficiently low to ensure consistent optical absorption patterns, reducing inter-subject variability.
5. In practical implementations, the wrist enables ease of attachment for wearable devices like bands or smartwatches, offering better user compliance than sites like the tongue or cheek, which are either invasive or difficult to access without medical assistance [55, 58, 62, 63].
6. Further in future research we can integrate the smart watch with BGM device, so it can serve all the purpose of smart watch and CGM altogether.

Hence, considering these clinical, anatomical, and engineering criteria, the wrist was selected as the sensor attachment site in this study. This choice ensures a balance between biological accessibility, signal quality, and user comfort, which are critical for developing a reliable, wearable BGM device.

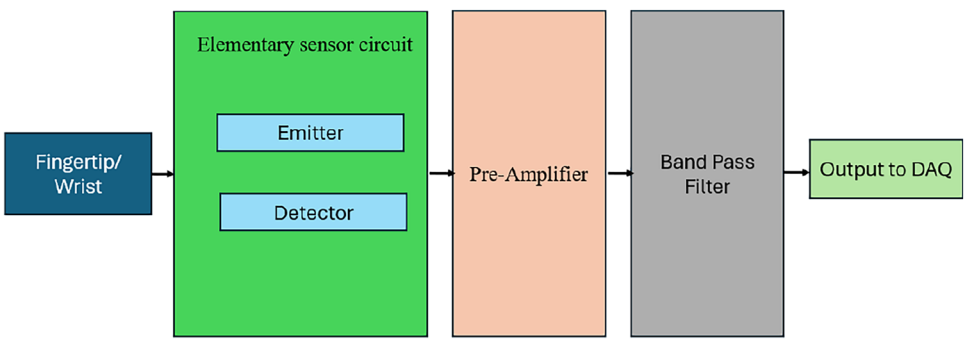


Figure 2. Basic architecture of sensor.

3.4 *Rationale behind sensor implementation*

The sensor's design and functionality were optimized based on the relationship between light wavelength and its ability to penetrate biological tissue. Glucose molecules in blood reside predominantly within the capillary networks of the dermal layer, situated roughly 2.0 mm beneath the skin surface–a depth well within the detection range of near-infrared (NIR) sensors [37]. Studies demonstrate that glucose absorption is most pronounced at specific wavelengths, including 660 nm, 940 nm, 1550 nm, and 1650 nm, with the greatest penetration capability observed at 940 nm [24, 28, 46, 61]. However, wavelengths below 700 nm and above 950 nm face limitations in tissue penetration due to intense absorption by hemoglobin and water. Penetration efficiency reaches its maximum in the 900–1000 nm spectrum before diminishing sharply beyond this range [28, 37, 52, 61].

Within the 940 nm spectrum, competing absorption effects from other biological constituents, including water, hem- oglobin, and melanin, are significantly reduced, establishing this wavelength as the most suitable choice for the study [21, 51, 64]. Following calibration and data processing, the resulting output correlates with blood glucose concentration levels. These measurements originate from voltage fluctuations triggered by shifts in optical density caused by the absorption process. Figure 3 provides a detailed schematic of the sensor's architecture.

3.5 *Hardware architecture of the device prototype*

The architecture of the non-invasive blood glucose monitoring system is shown in figure 4. The system utilizes the NodeMCU Esp8266 as the primary computing module, powered by a 3.3 V supply. Figure 4 illustrates the design of the blood glucose sensor for diabetic patients, which consists of multiple circuit blocks labeled A-E. In figure 4(A), the hardware of non-invasive IR sensor with band pass filter is demonstrated, where the IR sensor placed in front of wrist. The filtered transmitted signal by the sensor module called PPG signal as shown in figure 4(B) depends on the wavelength of light and the amount of radiation absorbed by the tissues [65, 66]. The NodeMCU Esp8266, shown in figure 4(C), converts the analog signal from figure 4(B) into a digital signal as shown in figure 4(D). Figure 4(E) displays the prototype of device whose signal is used to manage system operations such as data collection, information processing, and component control. Finally, the processed data from the microcontroller is analyzed on a personal computer for BG estimation and display, as depicted in figure 4(F).

The hardware architecture employs Short-Wave Near-Infr-ared (SW-NIR) technology, functioning across the 700–1300 nm spectrum. To identify blood glucose molecules, NIR sensor equipped with a 940 nm light source and a photodetector responsive to wavelengths from 900 to 1700 nm was chosen. This sensor demonstrates a sensitivity range of 0.9–0.95 A/W.

3.6 *Device operation and measurement methodology*

The system operates through the following steps: NIR sensor with filter interfaces with a microcontroller. Upon positioning the device prototype sensor against the user's skin, it captures readings and sends these signals to the microcontroller. The embedded processor converts the raw data into structured output, which is then transferred to a PC or mobile device via USB or wireless Bluetooth connectivity for additional evaluation. Glucose concentration in the blood is quantified via a spectroscopic technique utilizing an emitter and a photodetector. The photodetector captures attenuated light signals reflected back, which diminishes due to partial absorption by blood vessels in the wrist. To maintain precision, tests are performed in isolated settings to minimize background light contamination.

When incoming reflected light strikes the photodetector, it produces an electrical current. The detector's voltage output is digitized by the NodeMCU into discrete data points, which are transmitted to a spreadsheet on a connected laptop. Blood glucose levels are determined using an algorithm based on non-invasive data correlations, with final results visualized on the laptop interface. The system is powered by a 3.4 V DC source. To ensure stable operation, an LM7805 voltage regulator was utilized to deliver a steady 5V output to both the infrared LED and NodeMCU board. A resistor in series within the LED circuit maintains steady current flow, stabilizing light emission intensity. Key obstacles include precise quantification of tissue blood volume and protecting delicate circuitry from electromagnetic disturbances. These challenges can be addressed by implementing a gallium-indium-arsenide (GaInAs) photodiode, which operates across a spectral range that aligns with four separate LED wavelengths (450 nm, 900 nm, 1,350 nm, and 1,800 nm) [37]. Signal integrity is further improved by soldering connections between components. Unlike programmable digital systems with microprocessors, microcontrollers execute fixed routines, lacking the adaptability to install customized operating systems or user-selected software.

Figure 5 shows the schematic diagram of the BGML system with device prototype. A consistent testing protocol was established to account for variables affecting measurement reliability during prototype use. Healthy, non-diabetic volunteers were seated in a stationary position during trials. The device was secured to participants' wrists to stabilize peripheral blood flow dynamics influenced by posture shifts or localized pressure changes. Subjects rested their device-equipped arm on a support surface and used a hand warmer pre-measurement to optimize blood

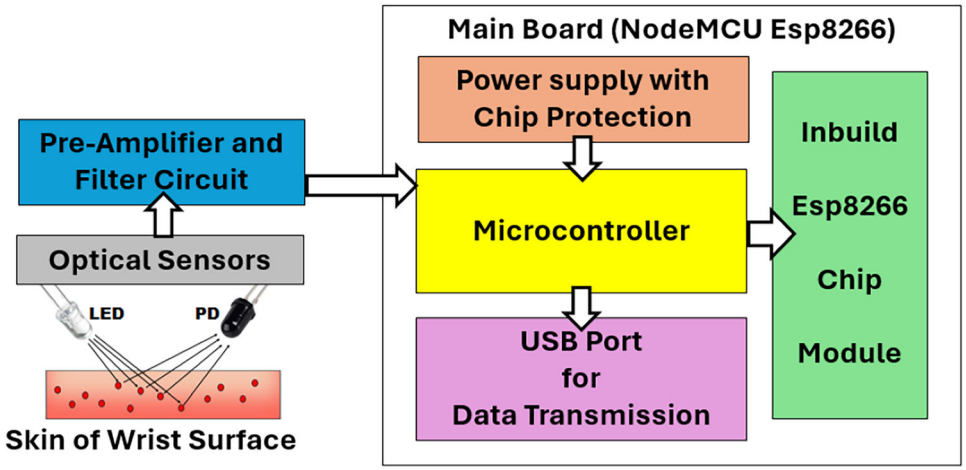


Figure 3. Structural diagram of an infrared sensor module.

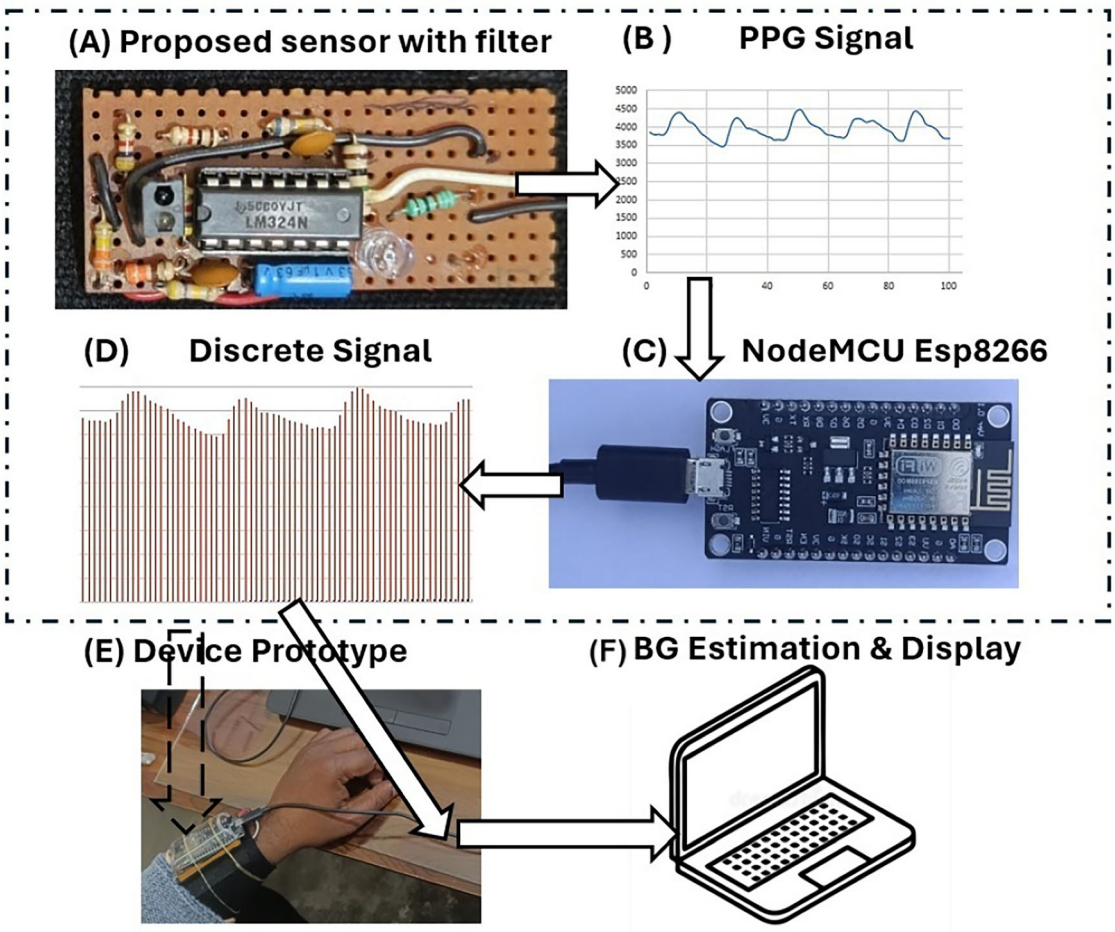


Figure 4. The architecture of BGLM system under the proposed method.

circulation. A Dr More-phon glucometer was used as the reference device for BGLMs. It is known that glucometers can exhibit deviations in readings even for the same blood sample. To avoid potential distortions in the PPG signal caused by body movements during prototype usage, no glucometer measurements were recorded during the PPG

signal acquisition period. Data recording began within 1 minute after the invasive glucometer measurement. The PPG signal was recorded for 90 seconds for each subject, capturing a total of 500 samples. A subset of 100 samples was used to calculate the AC and DC components of the PPG signal.

4. Blood glucose prediction using machine learning

To develop a reliable blood glucose prediction model, the AC component of the PPG signal is analyzed which carries the reflection of pulsatile blood flow and blood gulcose concentration. From each subject's PPG signal, the Root Mean Square (RMS) value of the AC component, denoted as AC_{rms}, was extracted after preprocessing and digital filtering. This value served as the independent variable (feature) in our regression models, while the corresponding invasively measured BGL using a Dr. Morephon glucometer served as the dependent variable (label). Using the collected dataset from 30 subjects, the 15 records were used to train the models. 5-fold cross-validation was applied to avoid overfitting. Both LR and SVM regression algorithms were implemented using scikit-learn library in Python. The models learned the linear relationship between AC_{rms} and BGL, based on minimizing the mean squared error (MSE) between predicted and reference BGL values.

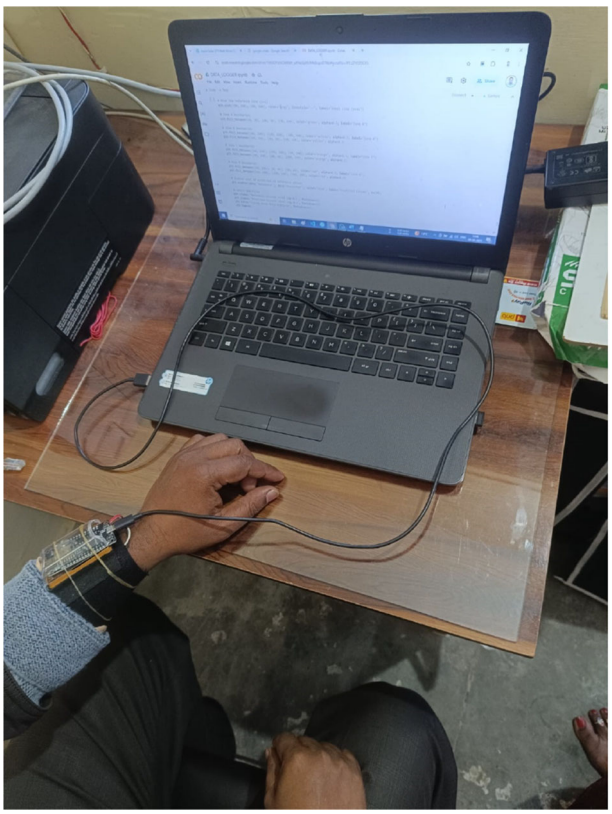


Figure 5. Non-invasive BGLM system with device prototype.

After training, Linear Regression model produces the predicted Blood Glucose Level (BGL_{LR}) equation as below:

$$BGL_{LR}(\text{mg/dL}) = -0.1180 \times AC_{rms} + 157.8354 \quad (7)$$

and similarly, the equation produced after training with SVM Regression is given by:

$$BGL_{SVM}(\text{mg/dL}) = -0.1074 \times AC_{rms} + 151.8019 \quad (8)$$

where AC_{rms} shows the rms value of AC component which carries the reflection of pulsatile blood flow and blood glucose concentration. The terms -0.1180 & 157.8354 from equation (7) and the term -0.1074 & 151.8019 from equation (8) are the coefficients which can be represented by a standard linear equation as:

$$y = mx + c \quad (9)$$

where m is the slope and c is the intercept. The negative slope i.e. -0.1180 in equation (7) and -0.1074 in equation (8) shows the inverse relationship between AC_{rms} and blood glucose concentration. It means that when AC_{rms} is higher, the blood glucose concentration shall be lower, and vice-versa.

The trained models were then validated using the remaining 15 subject records, and the predicted values were evaluated against reference measurements using RMSE, and R^2.

Performance metrics for accuracy evaluation: To evaluate the accuracy of the proposed method, the differences between the system's estimates and the reference glucometer values were analyzed using RMSE and R^2. The RMSE and R^2 are expressed as follows:-

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i - \hat{X}_i)^2} \quad (10)$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(X_i - \hat{X}_i)^2 \quad (11)$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|(X_i - \hat{X}_i)\right| \quad (12)$$

$$R^2 = 1 - \frac{(X_i - \hat{X}_i)}{(X_i - \bar{X}_i)} \quad (13)$$

Where X_i is the actual value, $\hat{X}_i$ is the predicted value, N is the total number of observations, and $\bar{X}_i$ is the mean of actual value of blood glucose level and calculated as follows:

$$\bar{X_i} = \frac{\sum X_i}{N} \tag{14}$$

Clarke Error Grid Analysis (CEGA) method assesses the clinical significance of differences between the BGLM system under the proposed method and the reference glucometer. The grid is divided into regions from A to E; A: Clinically accurate results, B: Clinically acceptable results, C: Unnecessary treatment, D: Failure to detect a dangerous condition, E: Misclassification of hypoglycemia as hyperglycemia or vice versa. For accuracy enhancement, the complete sample size of 500 was taken as compared to 100 samples for calculation of AC and DC value of PPG Signal. The LR and SVM model for generating BG prediction were used.

For further improvement, a unique method considering DC value having the information of physiological traits responsible for error have been proposed. In our unique method the ECF estimated by correspond to the DC value. The LR and SVM models are used for calculating ECF equations. Using these methods, the relationship between DC value and ECF is obtained in equation (15) and (16) as follows:

$$ECF_{LR}(mg/dL) = 0.0034 \times DC - 3.4269 \tag{15}$$

$$ECF_{SVM}(mg/dL) = 0.0037 \times DC - 4.7865 \tag{16}$$

So, the final equations for blood glucose level prediction (PBGL) corresponding the LR and SVM model is presented as follows:

$$PBGL_{LR}(mg/dL) = BGL_{LR} + ECF_{LR} \tag{17}$$

$$PBGL_{SVM}(mg/dL) = BGL_{SVM} + ECF_{SVM} \tag{18}$$

where ECF_{LR} and ECF_{SVM} denotes the estimated error correction factor using LR and SVM model respectively.

By using above method, after performance evaluation the results has been improved significantly and is discussed in next section.

5. Results and discussion

30 PPG signal records were evaluated to validate the proposed method. Figure 6 shows the predicted blood glucose levels using LR and SVM model along with reference glucose level.

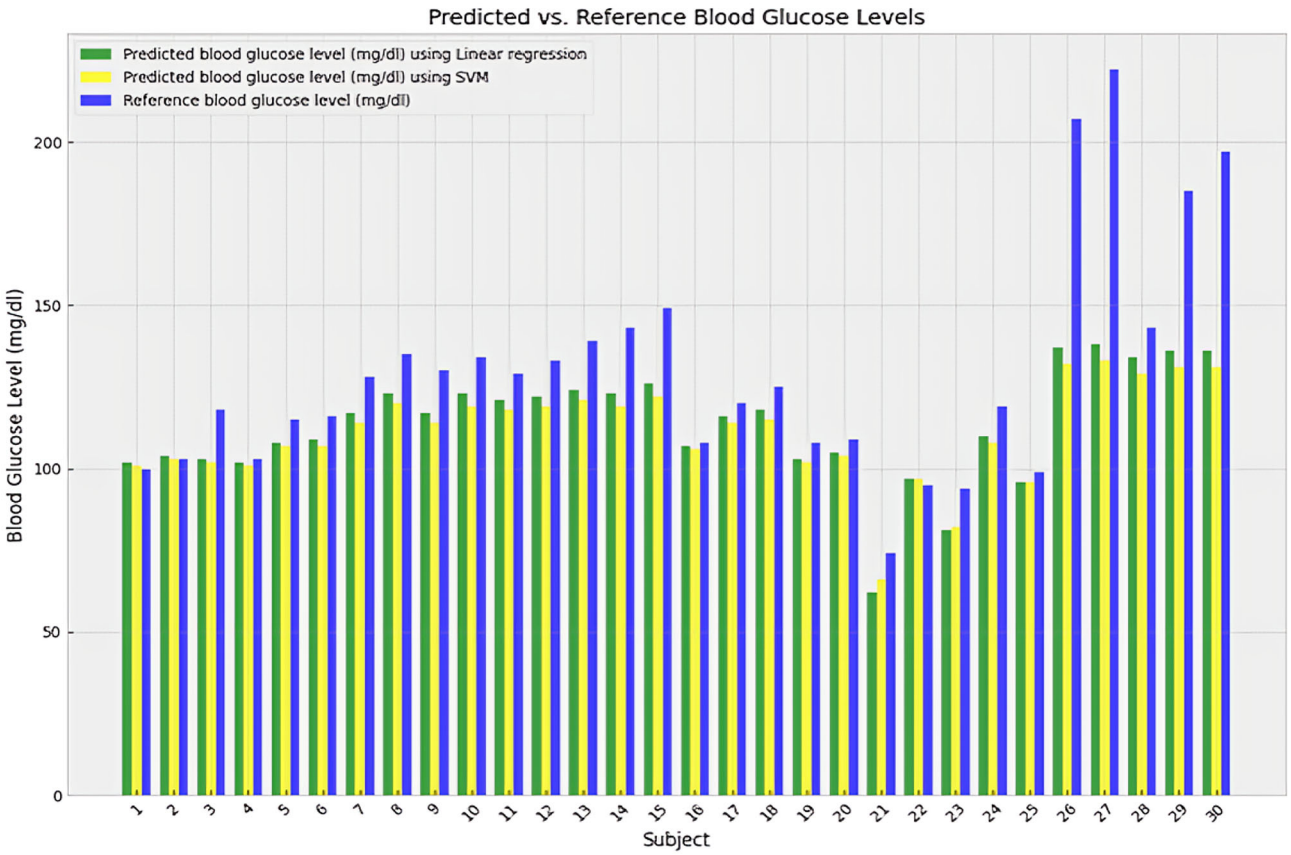


Figure 6. Reference and predicted blood glucose levels on 30 subjects in the validation of proposed method.

Table 1. Performance comparison with previous work.

Reference	No. of Samples	MAE (mg/dl)	MSE $(mg/dl)^2$	RMSE (mg/dl)	$\mathbf{R^2}$f Value	CEGA
[21]	50	N/A	N/A	N/A	0.94	A = 86.01%, B = 13.99%
[28]	12	N/A	N/A	N/A	0.937	No
[29]	19	N/A	342.9904	18.52	N/A	Yes
[36]	34	N/A	0.08	0.2828%	0.922	A = 20%, B = 20%
[37]	249	N/A	N/A	1720	0.578	Yes
[38]	52	N/A	289.6804	17.02	N/A	No
[39]	20	N/A	0.096	0.3098%	0.923	A = 90%, B = 10%
[40]	10	N/A	108.9936	10.44	0.839	Yes
[48]	36	N/A	55.6516	7.46	N/A	A = 100%
[49]	611	N/A	N/A	N/A	0.91	A = 80.6%, B = 17.4%
[64]	260	11.87	N/A	20.34	N/A	A = 87.89%, B = 12.11%
[67]	26	N/A	N/A	0.357%	N/A	No
Proposed method	50	4.6	26.60	5.16	0.958	Yes

N/A: Not available in the corresponding research work

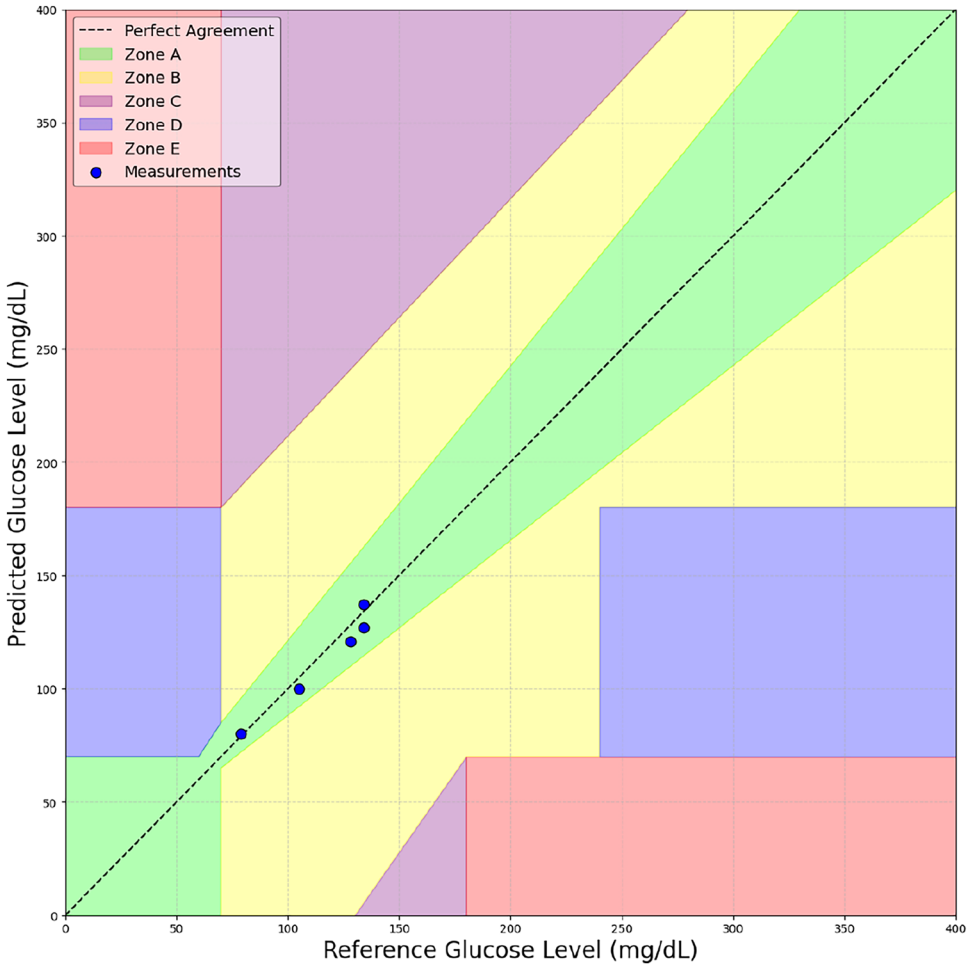


Figure 7. Clarke Error Grid Analysis of BGLM system under proposed method.

The size of 100 samples has been taken for calculation of AC and DC value of PPG Signal. Using LR, the corresponding value of RMSE & R^2 are 21.88 and 0.5766, respectively. Further using SVM the corresponding value of

Table 2. Blood glucose level.

Condition	Accu-Check Insta	Dr. Morephon	SRC Lab	SKA University	Proposed Model
Fasting	122	113	109	123	119

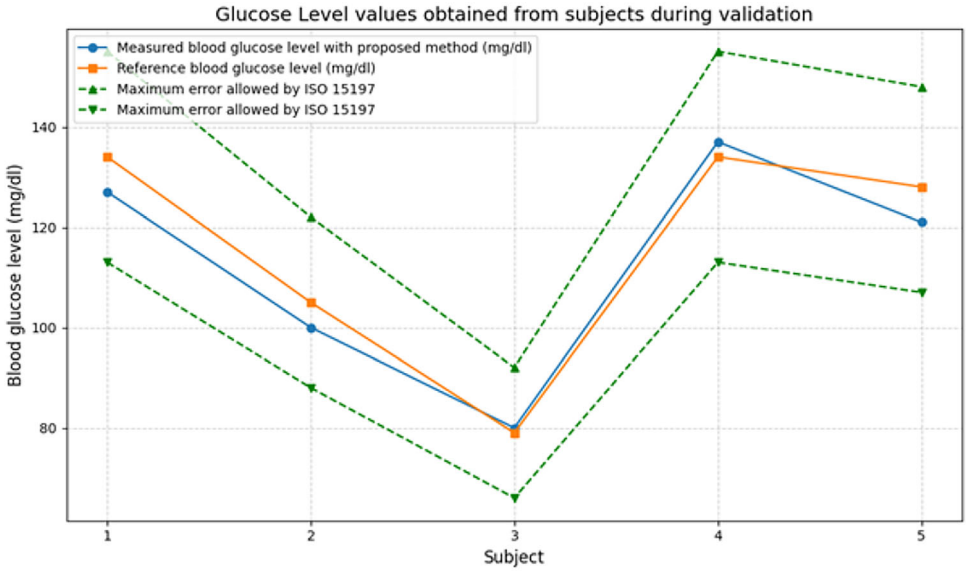


Figure 8. Glucose Level values obtained from subjects during validation.

RMSE & R^2 are 23.96 and 0.4924, respectively. The obtained values are similar to already existing work reported in literature [21, 28, 36, 38, 39, 50, 52]. To achieve more accuracy the complete sample size of 500 has been taken as compared to 100 samples for calculation of AC and DC value of ppg signal. Again, by using LR, the value of RMSE & R^2 are improved by 14.03% and 71.5%, respectively. The obtained values need to be improved significantly as compared to earlier work reported in literature [21, 28, 36, 38, 39, 50, 52].

For further improvement, a novel technique is used in proposed method by considering DC value corresponding to the BGL error. The DC value having the information of physiological traits that is responsible for error. The BGL prediction is done by using equations 15 and 16. The statistical analysis of the results indicates that the glucose level estimation using the SVM-based regression model achieves an RMSE of 5.16 mg/dl and an R^2 value of 0.958. This clearly indicate that a 48% reduction in RMSE and 13% improvement in R^2. While there is ongoing debate about the best method to evaluate the accuracy of glucose monitoring systems, RMSE is widely used to assess the overall performance of such systems.

However, many researchers consider R^2 to be the most reliable metric for demonstrating system performance, despite its inherent limitations, as highlighted in studies [21] and [36]. Table 1 provides a comparative analysis, showing improvements in both R^2 and RMSE compared to previous works.

Furthermore, Clarke Error Grid Analysis reveals that 99% of the estimates from the proposed device fall within Zone A, with the remaining 1% in Zone B, as illustrated in figure 7.

Table 2 show the deviation in the readings of the commercially available Invasive glucometers and the results of Lab reports for the same sample and time duration. The max difference is 14 mg/dl and the mean of all above readings is 117.2 mg/dl. Now if we assume that the reference value is 117 mg/dl then the accuracy is 98.29%.

5.1 *Compliance with international standards*

The current international standard for glucose measurement systems, ISO 15197:2015, specifies two key criteria for accuracy; (1) For blood glucose concentrations below 100 mg/dL, at least 95% of the estimated values must have an error margin of ±15 mg/dL compared to the reference method, and (2) For concentrations equal to or above 100 mg/dL, the acceptable error margin is ±15%. These criteria

ensure the reliability and precision of glucose monitoring devices. Figure 8 compares the glucose estimates generated by the proposed method with those derived solely from the NIR band, as suggested in earlier studies. The results are arranged in ascending order based on the reference glucose concentration.

Although manufacturers of glucose monitors typically do not report R^2 values, research studies have shown that commercial devices achieve R^2 values ranging from 0.578 to 0.94 ([21, 28, 29, 36, 38, 40, 41, 50, 52, 53, 64, 67]). The proposed method demonstrates comparable accuracy to commercial devices, with the added advantage of enabling non-invasive glucose estimation.

6. Conclusion and future scope

The BGLM system under the proposed method achieves the value of RSME and R^2 of 30.83% and 2.13%, respectively, which significantly improved compared to previous work. The Clarke Error Grid Analysis also confirms a 99% of prediction in clinically accurate zone A. The system under the proposed method is easy to use, less costly, less complex and most suitable for wearable real time BGM systems.

In future, large number of subjects may be considered for further improvement to its capability for universal and recalibration free BGLM device as compared to existing commercially available device like GlucoTrack, developed by the U.S.–Israeli company and CoG - Hybrid Glucometer, developed by Cnoga Israel.

List of symbols

ECF	Error Correction Factor
h	Planck's constant
I_r	Intensity of light
I_r	intensity of an LED
k	molar absorption coefficient
T_r	transmittance ratio
μ	Attenuation coefficient due to blood glucose
μ_s	Attenuation coefficients due other substances of blood and body

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